Chapter 9: The Hyperspatial Resolution of the Mott Problem

9.1 The Paradox of Localization

The Mott Problem questions why expanding quantum wave functions collapse into sharp linear tracks in bubble chambers. BMI theory resolves this by treating the electron as a quantized, compactified topological defect pinned within the **Thick Brane Interface (Σ)**, rather than a disembodied probability cloud.

9.2 The Screening Gate

Localization is governed by the **Screening Gate Function**, which modulates the interface permeability based on local curvature (R):

$$g_{eff}(R) = 1 / (1 + e^{-\alpha(R - R_{crit})})$$

When the wave encounters a target nucleus, the local curvature spike exceeds R_{crit}, causing the field to decouple from its spherical symmetry and "pin" itself to the coordinate of the target.

9.3 Topological Pinning

The system minimizes its total energy functional by routing the diffuse tension of the spherical wave into a localized exponential bound state. This is a causal, mechanical process of energy minimization, not an abstract wave-function collapse.

9.4 The Geodesic Wake

The linear track is the "geodesic wake" of a hyper-dimensional soliton. Once pinned, the electron's directional shear is locked. It cannot re-expand; it must propagate linearly, sequentially "triggering" subsequent ionizations as it traverses the chamber. This confirms that bubble chamber tracks are the visual history of a localized crease propagating through the manifold.